

CSCE 669 COMPUTATIONAL OPTIMIZATION · MODULE 04 · LECTURE NOTES

Duality

Every minimisation has a shadow maximisation.

MODULE OUTCOMES

- Construct the Lagrangian and the dual function.
- State weak and strong duality and the conditions for each.
- Interpret dual variables as shadow prices.
- Apply complementary slackness.
- Use the dual as a certificate of optimality.

1 Constructing the dual

$$L(\mathbf{x}, \lambda, \mathbf{v}) = f(\mathbf{x}) + \sum_i \lambda_i g_i(\mathbf{x}) + \sum_j v_j h_j(\mathbf{x}) \rightarrow g(\lambda, \mathbf{v}) = \inf_{\mathbf{x}} L(\mathbf{x}, \lambda, \mathbf{v})$$

The Lagrangian prices each constraint violation and adds it to the objective, with $\lambda \geq 0$ for inequalities and v unrestricted for equalities. The dual function is the infimum of the Lagrangian over $\mathbf{x}$ — and because it is a pointwise infimum of functions that are *affine* in $(\lambda, \mathbf{v})$, it is **concave, always, regardless of whether the primal problem was convex** (Module 02 §2's preservation rules, applied to the minimum of affine functions). So the dual problem — maximise g subject to $\lambda \geq 0$ — is a **convex problem no matter what the primal was**, which is precisely why duality is useful for hard problems and not only for easy ones.

WEAK DUALITY ALWAYS; STRONG DUALITY USUALLY

Weak duality: $g(\lambda, \mathbf{v}) \leq p^*$ for every dual feasible $(\lambda, \mathbf{v})$. This holds for any optimisation problem whatsoever — convex, non-convex, integer, combinatorial. The proof is two lines.

So any dual feasible point gives a lower bound on the primal optimum, for free. That is the single most useful consequence of duality, and it is what branch and bound prunes with (Module 09) and what approximation guarantees are measured against (Module 10).

Strong duality: the gap is exactly zero, $d^* = p^*$. For convex problems it holds under a mild regularity condition — Slater's condition: there exists a strictly feasible point, one satisfying every inequality constraint strictly.

So for a convex problem, solving either problem solves both, and a dual solution *proves* that a primal solution is optimal. A lower bound together with a matching feasible point is a certificate — which converts 'I could not find anything better' into 'nothing better exists', and that is a categorically different kind of claim.

2 What the dual variables mean

DUAL VARIABLES ARE PRICES

λ_i is the rate at which the optimal value improves if constraint i is relaxed by one unit — formally, $\partial p^*/\partial b_i = -\lambda_i$ where the problem is differentiable in the right-hand side.

So the dual variable *prices* the constraint. 'An additional hour of machine time is worth thirty-four pounds' is a dual variable, reported in the problem's own vocabulary rather than in mathematical notation — **and stating it that way is what makes an optimisation result actionable to the person who owns the problem.**

A zero dual variable means the constraint is not binding. Relaxing it buys nothing because you are not currently pressed against it, so no investment in that resource is worthwhile.

This is what makes duality practically valuable rather than merely formally elegant. It tells the problem owner where to invest — which constraint to buy more of, which supplier to negotiate with, which deadline to extend — **and the primal solution alone does not.** Project 1 requires exactly this interpretation, and it is the part that converts a solved model into a decision.

$$\lambda_i \cdot g_i(x^*) = 0 \quad \text{for every } i \quad (\text{complementary slackness})$$

At optimality, for each constraint, at least one of the dual variable and the constraint slack is zero. **A slack constraint has zero price** — it is not limiting you, so relaxing it is worthless. **A positive price implies the constraint is tight** — you are pressed against it, which is why it has value. **This is the economic statement of the obvious: you do not pay for something you are not using**, and framing it that way makes it memorable in a way the algebra does not.

3 What duality is actually for

Use	How	Where it appears
Certifying optimality	A primal feasible point and a dual feasible point with matching values prove each other optimal.	Every solver reports this gap, and it is how you know a solution is finished rather than merely stalled.
Bounding a hard problem	Any dual feasible point is a valid lower bound, by weak duality, even when the primal is intractable.	Branch and bound (Module 09 §2) prunes entirely on these bounds.
Sensitivity analysis	Shadow prices on every constraint (§2).	The main practical payoff in applied work, and the thing clients actually ask for.
Solving an easier problem	The dual is sometimes structurally simpler than the primal.	The SVM: the dual has one variable per example and enables kernels (CSCE 633 Module 07) — see the callout.
Decomposition	Split a problem coupled by a few shared constraints by pricing the coupling and solving the parts independently.	Large-scale structured problems; Dantzig–Wolfe and Benders decomposition.
Theory	Max-flow min-cut, von Neumann's minimax theorem, and LP duality are the same theorem.	Several famous results that look unrelated turn out to be instances of one — which is genuinely satisfying and worth noticing.

THE SVM DUAL IS WHY KERNELS EXIST

The support vector machine's primal problem optimises over the weight vector w , whose dimension equals the number of features — which is a problem when the feature space is infinite-dimensional.

Its dual optimises over one variable per training example, and the data enters the dual only through inner products $x_i^T x_j$ — individual feature vectors appear nowhere.

That is exactly what makes the kernel trick possible (CSCE 633 Module 07 §2). You can substitute a kernel function for the inner product *only because* the dual formulation never requires the vectors themselves — which is why that course could not fully justify the trick and this one can.

So kernels are a consequence of taking the dual, not an independent idea, and the representer theorem is the same observation stated differently (CSCE 633 Module 07 §2) — the solution lies in the span of the data because the dual has one variable per datum. **Three apparently separate results, one mechanism.**

4 When strong duality fails

THE DUALITY GAP IS WHERE THE DIFFICULTY LIVES

For non-convex problems strong duality generally fails: the dual optimum is strictly below the primal optimum, and the difference is the **duality gap**.

That gap is a quantitative measure of the problem's non-convexity — and it is precisely what makes the problem hard. A problem with zero gap can be solved through its dual; a problem with a large gap cannot.

The dual bound remains useful regardless. Weak duality still holds, so the dual optimum is still a valid lower bound — **which is exactly what branch and bound prunes with** (Module 09 §2), and the tighter the bound the smaller the search tree.

And the integrality gap of Module 10 is this gap, specialised to integer programs and their linear relaxations. **So approximation guarantees are duality gaps with a ratio attached** — which unifies Modules 04, 09, and 10 under a single idea, and is the clearest reason to learn duality properly rather than treating it as a formality.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Boyd & Vandenberghe — Convex Optimization, chapter 5 (free)	The definitive treatment of this entire module , including the sensitivity interpretation and Slater's condition.
Boyd — EE364A lectures on duality (free video)	The geometric intuition for weak and strong duality, which the algebra alone does not convey.
MIT 15.093 — LP duality and sensitivity analysis (free)	The \$2 shadow price material , with worked economic interpretations.
Vanderbei — Linear Programming: Foundations and Extensions (free PDF)	Free, and strong on duality's practical use and on the game theory connection in §3.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Derive the dual** of a linear program by hand, from the Lagrangian.
- 2 **Verify weak duality numerically** on a non-convex problem by evaluating the dual at several feasible points.
- 3 Solve a convex problem and its dual and confirm the values match.

- 4 **Construct a problem violating Slater's condition** and show a nonzero duality gap.
- 5 **Interpret every dual variable** of your project model in the problem's own vocabulary.
- 6 **Verify complementary slackness** on a solved instance, constraint by constraint.
- 7 **Perturb a binding constraint by one unit** and confirm the objective changes by the dual variable.
- 8 Derive the SVM dual and identify where the inner products appear.
- 9 **Show max-flow min-cut is LP duality** by writing both programs.
- 10 Use a dual bound to prune a small branch-and-bound search by hand.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Construct the Lagrangian and the dual function, and say why the dual is always convex.
- 2 State weak duality and say when it holds.
- 3 State strong duality and the condition for it.
- 4 What do dual variables mean, and why does that matter practically?
- 5 State complementary slackness and give its economic reading.
- 6 Name six uses of duality.
- 7 Why does the SVM dual enable kernels?
- 8 What is the duality gap, and what does it measure?